

Universidad Rafael Landivar

Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y telecomunicaciones - sección 1

Catedrático: Ing. Carlos René Portillo Sen

REDES Y TELECOMUNICACIONES:

Parcial#1

Rodrigo José Ruiz Juez: 1037623

Guatemala, 29 de febrero de 2026

Enlace a video:

<https://youtu.be/0KZlc83quKA>

Excell

VLAN	Departamento	Hosts Actuales	Hosts +30%	Hosts Soportados	Subred Asignada	Prefijo	Máscara	Gateway	Rango Utilizable	Broadcast
10	Operaciones	80	104	126	10.0.0.0	/25	255.255.255.128	10.0.0.1	10.0.0.2 - 10.0.0.126	10.0.0.127
20	Ventas	60	78	126	10.0.0.128	/25	255.255.255.128	10.0.0.129	10.0.0.130 - 10.0.0.254	10.0.0.255
30	TI	45	59	62	10.0.1.0	/26	255.255.255.192	10.0.1.1	10.0.1.2 - 10.0.1.62	10.0.1.63
50	VoIP	30	39	62	10.0.1.64	/26	255.255.255.192	10.0.1.65	10.0.1.66 - 10.0.1.126	10.0.1.127
40	RRHH	22	29	30	10.0.1.128	/27	255.255.255.224	10.0.1.129	10.0.1.130 - 10.0.1.158	10.0.1.159
60	Gerencia	10	13	14	10.0.1.160	/28	255.255.255.240	10.0.1.161	10.0.1.162 - 10.0.1.174	10.0.1.175
70	Servidores	6	8	14	10.0.1.176	/28	255.255.255.240	10.0.1.177	10.0.1.178 - 10.0.1.190	10.0.1.191
99	Gestión	0	10	14	10.0.1.192	/28	255.255.255.240	10.0.1.193	10.0.1.194 - 10.0.1.206	10.0.1.207

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User Access Verification

Password:

SW-Core1>enable

Password:

SW-Core1#show vtp status

VTP Version capable : 1 to 2
VTP version running : 2
VTP Domain Name : TECHCORP
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 0001.C922.E700

Configuration last modified by 0.0.0.0 at 3-2-93 20:39:45

Local updater ID is 10.0.0.1 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :

VTP Operating Mode : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 14
Configuration Revision : 37
MD5 digest : 0x7A 0x62 0x97 0x6D 0xCC 0xAA 0x95 0x53
0x98 0x22 0x5D 0x4D 0x91 0xA1 0x95 0xDC

SW-Core1#show vtp cont

SW-Core1#show vtp counters

VTP statistics:

Summary advertisements received : 644
Subset advertisements received : 78
Request advertisements received : 63
Summary advertisements transmitted : 177
Subset advertisements transmitted : 82
Request advertisements transmitted : 0
Number of config revision errors : 0
Number of config digest errors : 299
Number of V1 summary errors : 0

VTP pruning statistics:

Trunk Join Transmitted Join Received Summary advts received from
non-pruning-capable device

SW-Core1#

SW-Core1#

SW-Core1#

SW-Core1#

SW-Core1#

VTP pruning statistics:

Trunk	Join Transmitted	Join Received	Summary advts received from non-pruning-capable device
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```
SW-Core1#
SW-Core1#
SW-Core1#
SW-Core1#
SW-Core1#
SW-Core1#
SW-Core1#show
SW-Core1#show in
SW-Core1#show interfaces tr
SW-Core1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    999
Po2       on        802.1q         trunking    999

Port      Vlans allowed on trunk
Po1       10,20,30,40,50,60,70,99,999
Po2       10,20,30,40,50,60,70,99,999

Port      Vlans allowed and active in management domain
Po1       10,20,30,40,50,60,70,99,999
Po2       10,20,30,40,50,60,70,99,999

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,40,50,60,70,99,999
Po2       10,20,30,40,50,60,70,99,999
```

```
SW-Core1#
SW-Core1#show et
SW-Core1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

Number of channel-groups in use: 2
Number of aggregators: 2

Group	Port-channel	Protocol	Ports
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Number of aggregators: 2

Group	Port-channel	Protocol	Ports
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1	Po1(SU)	LACP	Gig0/1(P) Gig0/2(P)
2	Po2(SU)	LACP	Fa0/1(P) Fa0/2(P)

SW-Core1#spa

SW-Core1#spanning-tree vlan 10

% Invalid input detected at '^' marker.

SW-Core1#show

SW-Core1#show sp

SW-Core1#show spanning-tree vlan

SW-Core1#show spanning-tree vlan 10

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 4106
Address 000A.41DE.1647
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4106 (priority 4096 sys-id-ext 10)
Address 000A.41DE.1647
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Po1	Desg	FWD	3	128.27	P2p
Po2	Desg	FWD	12	128.28	P2p

SW-Core1#show spanning-tree vlan 30

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 4126
Address 0030.A363.66C4
Cost 3
Port 27 (Port-channel1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)
Address 000A.41DE.1647
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Po1	Root	FWD	3	128.27	P2p
Po2	Desg	FWD	12	128.28	P2p

SW-Core1#show ip ro

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```

-----
Po1          Root FWD 3          128.27   P2p
Po2          Desg FWD 12         128.28   P2p

SW-Core1#show ip ro
SW-Core1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.0.0.0/25 is directly connected, Vlan10
C       10.0.0.128/25 is directly connected, Vlan20
C       10.0.1.128/27 is directly connected, Vlan40
C       10.0.1.160/28 is directly connected, Vlan60

SW-Core1#show
SW-Core1#show ip
SW-Core1#show ipinter
SW-Core1#show ip inter
SW-Core1#show ip interface bri
SW-Core1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Port-channel1	unassigned	YES	unset	up	up
Port-channel2	unassigned	YES	unset	up	up
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down

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PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.1.10

Pinging 10.0.1.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Request timed out.

Ping statistics for 10.0.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.1.10

Pinging 10.0.1.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Request timed out.
Reply from 10.0.0.1: Destination host unreachable.

Ping statistics for 10.0.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=128
Reply from 10.0.0.10: bytes=32 time=4ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time=4ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>ping 10.0.0.140

Pinging 10.0.0.140 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.140: bytes=32 time<1ms TTL=127
Reply from 10.0.0.140: bytes=32 time<1ms TTL=127
Reply from 10.0.0.140: bytes=32 time=1ms TTL=127

Ping statistics for 10.0.0.140:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.0.1.10
```

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